

Owmee E2E Product and Operations Flow Design

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1. Scope

This document describes the complete Owmee end-to-end flow across buyer, seller, FE, admin, finance, warehouse, and provider systems. It covers happy paths, negative paths, state transitions, and launch-critical edge cases.

Non-goals:

- Buyer/seller chat.
- Buyer/seller meetup.
- Free-form Make Offer notes.
- Offline FE cash/UPI collection.

2. Actors

Actor	Responsibilities
Buyer	Browse, wishlist, offer/buy, pay, track, accept delivery, request return/dispute
Seller	Create listing, confirm listing facts, respond to offer, confirm readiness, complete payout/KYC when needed
FE	Visit seller, verify seller, inspect item, capture proof, deliver/return when assigned
Ops admin	Assign FE, route hub items, monitor stuck workflows, handle exceptions
Risk/admin	Review fraud decisions, reports, disputes, blocks
Finance	Release/refund/payout queues, direct acquisition payout ledger
Warehouse/admin	Receive direct acquisition custody, quarantine mismatch, approve buyer-facing inventory
Providers	MSG91, Razorpay, Gemini, Bureau, KYC, R2, geo, FCM/noop

3. Flow A - User Onboarding

User enters phone
-> API sends OTP through configured SMS adapter
-> User verifies OTP
-> API creates/updates user/session/device
-> Address gate checks saved address state
-> User can browse, list, offer, or buy
-> Fraud check may run in background

Positive cases:

- Returning user with saved address goes directly into the app.
- New user without address is sent through address capture.
- OTP send failure deletes stored OTP so an undelivered code cannot be used.

Negative cases:

- Invalid OTP fails.
- Expired OTP fails.

- MSG91 delivery failure returns OTP_DELIVERY_FAILED.
- Production OTP whitelist cannot bypass MSG91 login.
- Risk decision can step up or block sensitive actions later.

4. Flow B - Address And Location

User grants location or searches manually

- > reverse geocode provider returns structured address
- > app validates address shape
- > user confirms address details
- > backend stores user address

Important rules:

- Address responses must be shaped and retryable.
- Backend failures must propagate so the app can show recovery.
- Transaction logistics uses immutable snapshots, not mutable current address.

5. Flow C - AI Listing Creation

Seller chooses sell flow

- > uploads 3+ photos
- > direct R2 upload session is created
- > worker analyzes photos with Gemini
- > draft becomes ready
- > seller reviews buyer-visible facts
- > seller completes required P0 checks
- > backend validates same requirement contract
- > listing publishes or enters moderation as policy requires

P0 fields:

- Photos.
- Category.
- Title.
- Price.
- Condition.
- Defect/missing-parts/safety disclosure.
- Locality and pickup/delivery method.
- Category-specific fields.
- Working or battery status where relevant.

Category-specific gates:

- Toys: age, hygiene, parts, safety, checklist, battery/working status when powered.
- Books: type, language, page condition, markings, completeness, set/class/ board/edition where needed.
- Appliances: appliance type, working status, accessories, defects, pickup complexity for bulky/install-heavy items.
- Devices: brand/model/storage/RAM/identifier/accessory and condition facts.

Negative cases:

- Stock/catalog-only image blocks or forces review.
- Multiple unrelated products blocks or forces review.
- Private info, faces, cards, chats, UPI IDs, documents are safety blockers.
- Generic titles such as "Other Pink" are not acceptable.
- Vague negative disclosures need buyer-facing detail.

- Low-confidence P0 values require seller confirmation.

6. Flow D - Listing Moderation And Seller Inventory

Listing created

- > status active or moderation queue
- > seller inventory reads listing plus latest transaction state
- > buyer feed reads active buyer-facing listings

Seller inventory labels:

- Active: seller can manage listing.
- Payment pending: listing hidden/reserved while buyer payment attempt exists.
- Seller confirmation required: captured payment needs readiness.
- Pickup pending: seller confirmed readiness and pickup can be assigned.
- Sold/completed: show sold timestamp and order history.

Negative cases:

- Payment failure/cancel/timeout returns listing to available when safe.
- Captured payment must keep listing hidden/reserved.
- Lazy repair scopes to the seller and rolls back on failure.

7. Flow E - Browse, Search, Wishlist

Buyer opens home/search

- > API returns feed/listing cards
- > buyer opens listing detail
- > buyer can wishlist, offer, or buy

Rules:

- Listing detail must not call expensive providers.
- Trust labels must be precise: phone verified, identity verified, Owmee reviewed, seller confirmed, FE inspected.
- No "unverified" negative framing.
- No buyer-seller chat entry points.

8. Flow F - Make Offer

Buyer makes one active offer

- > seller accepts, rejects, or counters
- > buyer can update own price up to 3 times before seller response
- > counter can be accepted or rejected
- > accepted offer leads to transaction/payment

Rules:

- One active offer per buyer/listing.
- Seller counter expires after 48 hours.
- No re-counter.
- Reject/expiry can trigger buyer cooldown.
- Offer history is the structured communication record.

Negative cases:

- Duplicate active offer is rejected.
- Update limit is enforced.
- Counter expiry is enforced at accept time.
- Invalid status transitions are rejected.

9. Flow G - Buy Now And Payment

Buyer taps Buy Now

- > backend creates offer/reservation/transaction/payment attempt
- > listing becomes reserved/payment_pending
- > mobile opens Razorpay in-app checkout
- > success path verifies signature, order, amount, and captured status
- > Razorpay webhook can also capture the payment
- > transaction moves to payment_captured

Positive cases:

- Client confirmation verifies Razorpay signature.
- Webhook is signed and idempotent.
- Amount must match expected transaction amount.
- Already captured transaction returns idempotent success.

Negative cases:

- Invalid signature rejected.
- Underpayment rejected.
- Authorized but not captured waits for webhook.
- Payment failed releases inventory.
- Duplicate failure is idempotent.
- Cancelled unpaid transaction releases inventory only for owner.
- Late capture after cancel does not reopen sale.
- Capture after payment window expires releases and refunds.
- Timeout job releases stale unpaid attempts.

10. Flow H - Seller Readiness

Payment captured

- > seller readiness task opens
- > seller confirms item available, condition unchanged, accessories, pickup address, and pickup slot/readiness
- > admin can assign pickup

Rules:

- Readiness is structured, not chat.
- Admin pickup assignment requires readiness confirmed.
- Seller decline cancels/suppresses listing and refunds buyer.
- Missed readiness deadline goes to expiry/refund/ops review.

Negative cases:

- Payment captured without buyer address holds ops flow.
- Seller readiness confirm mutates only structured state.
- Seller decline triggers refund path and support evidence.

11. Flow I - Admin Pickup Assignment

Admin pickup queue

- > assign eligible FE
- > FE sees assigned pickup
- > FE starts route

Rules:

- FE must be active, trained, verified, device-approved where required, and certified for the category.
- Suspended or terminal FE profiles cannot receive work.
- Rebound devices deactivate FE until admin reapproval.

12. Flow J - FE Pickup And Inspection

FE reaches seller

- > verifies seller/arrival where applicable
- > captures item proof
- > checks item against listing snapshot
- > pass moves item to hub custody
- > fail triggers refund/rejection path

FE proof examples:

- Pickup photos.
- Condition notes.
- Accessories checklist.
- IMEI/serial for applicable devices.
- Safety and missing-parts checks for toys/books/appliances.

Negative cases:

- FE cannot complete pickup before assignment.
- FE cannot skip seller verification where required.
- Pickup fail refunds buyer and preserves evidence.
- Payout activity refuses release before pickup confirmation.

13. Flow K - Hub Routing

Item at hub

- > admin routes to FE delivery or courier
- > FE delivery assigned or courier AWB/status entered

Launch rule:

- Admin manually chooses route during pilot.
- Courier integration can be stub/manual while real integration is pending.
- Admin routing decision should be audited.

14. Flow L - Delivery And Buyer Acceptance

FE/courier delivers item

- > delivery proof captured
- > buyer tracking timeline updates
- > buyer accepts or window expires
- > payout can become release-eligible if seller payout/KYC checks pass

Rules:

- Delivery completion still holds payout until buyer confirmation or window expiry/dispute resolution.
- Buyer acceptance is separate from FE delivery proof.
- Buyer can request return/dispute through structured flow.

15. Flow M - Refunds, Returns, And Disputes

Buyer initiates issue

- > KYC/risk policy may require step-up
- > evidence upload requested
- > admin/risk reviews

-> refund/return/dispute decision updates transaction and payout path

Rules:

- Buyer KYC is invoked for refund/return/dispute escalation.
- Dispute evidence comes from structured records, FE proof, delivery proof, snapshots, and payment events.
- No chat archive exists because chat is not part of Owmee.

Negative cases:

- Return without evidence can be held or rejected.
- Duplicate refund must be idempotent.
- Refund retries must not double refund.
- Dispute resolution must move money through the shared payment/refund path.

16. Flow N - Seller Payout

Pickup passes

-> payout processing can start
-> seller payout/KYC status checked
-> delivery and acceptance/dispute window complete
-> payout release can occur

Rules:

- Payment capture alone does not release payout.
- Pickup pass starts processing only after Owmee validates custody.
- Unverified seller is prompted to complete payout verification.
- payout_released_at stays empty until all gates pass.

17. Flow O - Owmee Direct Acquisition

Seller accepts Owmee Direct offer

-> seller books address/slot and manifest
-> ops assigns FE
-> FE starts, arrives, verifies seller OTP/QR
-> FE item photos and QC
-> revise/reject/accept per item
-> admin approval needed for high price increase
-> seller final OTP acceptance
-> FE requests payout-ready
-> finance posts seller ledger credit
-> warehouse receives custody
-> admin approves buyer-facing listing

Negative cases:

- No offline booking.
- No QC before arrival verification.
- No price revision without evidence.
- No >10 percent price increase without admin approval.
- No payout for rejected items.
- No seller final acceptance while item QC is pending.
- FE cannot post payout.
- FE cannot complete warehouse handover.
- Warehouse mismatch quarantines payable items.
- Buyer-facing listing requires warehouse inbound and linked draft listing.

18. Flow P - FE Onboarding

Admin creates/invites FE

- > FE logs in through phone OTP
- > FE binds device
- > Admin completes verification, training, category certification
- > FE checks in to shift
- > FE can receive work only when ready

Blocks:

- Candidate FE cannot be assigned.
- Wrong category certification blocks assignment.
- Device rebind deactivates until admin reapproves.
- Suspended FE cannot shift or receive work.
- Terminal FE cannot bind device.

19. Flow Q - Notifications

Notification buckets stay transaction-oriented:

- Payment captured.
- Seller readiness needed/confirmed/declined.
- Pickup assigned/completed/failed.
- Hub routed.
- Delivery assigned/completed.
- Refund/return/dispute updates.
- Payout verification needed/released.

No notification should invite buyer/seller free-form messaging.

20. State Machine Summary

Stage	Typical state	Owner	Next positive transition	Failure transition
Listing	active	Seller/system	reserved	withdrawn/rejected
Payment	payment_pending	Buyer/Razorpay	payment_captured	payment_failed/cancelled/expired
Readiness	seller_readiness_pending	Seller	seller_ready	seller_declined/expired_refund
Pickup	pickup_assigned	Admin/FE	pickup_completed	pickup_rejected_refund
Hub	at_hub	Admin	delivery_assigned/courier_routed	ops_hold
Delivery	delivery_in_progress	FE/courier	delivered	delivery_issue
Acceptance	delivered	Buyer/system	completed	return/dispute
Payout	payout_processing	Finance/provider	payout_released	payout_blocked/manual_review

21. Core P0 Regression Checklist

- OTP login works with MSG91 or mock mode as configured.
- New user address gate blocks until valid address exists.
- AI listing draft can be created through async R2 upload.

- Seller review blocks missing P0 category requirements.
- Listing publish stores seller review snapshot.
- Buyer can browse active listing.
- Offer rules enforce duplicate/update/counter limits.
- Buy-now reserves listing and creates payment attempt.
- Razorpay success verifies signature and amount.
- Payment failure/cancel/timeout releases inventory.
- Captured payment opens seller readiness, not pickup directly.
- Seller decline refunds and suppresses listing.
- Admin cannot assign pickup before readiness.
- FE pickup pass/fail updates transaction correctly.
- Payout is not released before pickup and delivery gates.
- Buyer tracking timeline reflects payment/readiness/pickup/delivery.
- Return/dispute path requires structured evidence and policy.
- Direct acquisition cannot bypass FE/finance/warehouse/admin gates.
- FE onboarding blocks unready/suspended/wrong-category FE assignment.

22. Current Test Coverage Anchors

Area	Test files
Order/logistics contracts	test_order_e2e_contract.py
Checkout/payment failure/timeout	test_checkout_payment_flow.py
Direct acquisition functional gates	test_direct_acquisition_functional.py, test_direct_acquisition_flow.py
FE onboarding gates	test_fe_onboarding.py
Category requirements	test_listing_category_requirements.py
AI prompt safety/performance	test_ai_prompt_contracts.py, test_ai_vision_strategy.py
Verification policy	test_verification_flow_contracts.py, test_verification_policy.py
MSG91 adapter	test_sms_msg91_adapter.py, test_msg91_probe_script.py

23. Open Launch Risks To Track

- Courier provider integration can remain manual for pilot, but SLA and AWB ownership must be explicit.
- Real payment settlement/payout provider reconciliation needs finance runbook.
- FE fraud controls should be monitored with geofence, device binding, evidence, issue reporting, and admin audit trails.
- Legacy design-system allow-list still exists in mobile; new screens must not add drift.
- AI output quality should be monitored by prompt version, category, correction rate, and seller edit rate.